PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

ARUSHA REGION

FORM TWO - MOCK EXAMINATION

032 **CHEMISTRY**

Time: 2:30 Hours MARCH 2026

INSTRUCTIONS

1. This paper consists of section A, B and C
2. Section A carries fifteen (15) marks, section B carries seventy (70) marks and section C carries fifteen (15) marks.
3. All writing must be in blue or black ink EXCEPT drawings which must be in pencil.
4. All communication devices and any unauthorized material are not allowed in the assessment room.
5. Write your examination number at the top right Conner of every page.
6. The following atomic masses may be used:
   Na= 23, Ca = 40, Cl= 35.5, O = 16, N= 14, H= I

NO.	SCORE	SIGNATURE
1.		
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MOCK EXAMINATIONS ARUSHA

9.		
10.		
TOTAL		
CHECKER'S SIGN		

SECTION A(15marks)

Answer all questions from this section

1. For each of the item (i) – (x), choose the correct answer from the given alternatives and write its letter in the box provided.

(i) Your young brother was given a syrup with a prescription that it should be well shaken before use. This is because the syrup is a

(A) Suspension
(B) Clear solution
(C) Filtrate
(D) Homogeneous mixture

(ii) Which of the following is a compound?

(A) Helium (He)
(B) Hydrogen (H_2)
(C) Carbon monoxide (CO)
(D) Neon (Ne)

(iii) Fainting is different from shock because fainting is a sudden loss of.

(A) Weight of the body
(B) Consciousness
(C) Confidence in mind
(D) Water in the body

(iv) A form one student saw a flammable sign on a box. This suggest that the box is most likely to contain

(A) Fire wood
(B) Papers
(C) Radioactive material
(D) Spirit used in lamps

(v) Silver is one among the elements which their chemical symbols are derived from Latin name. Which alternative has a correct Latin name and chemical symbol of silver?

(A) Aurum (AU)
(B) Argentum (Ag)
(C) Argenteum
(D) Hydrargyrum (Hg)

(vi) Which aamong the following chemical substances does not fit in agricultural activities.

(A) Fabrics
(B) Fertilizer
(C) Pest icicles
(D) Vaccines

MOCK EXAMINATIONS ARUSHA

(vii) A student who gets burnt accidentally in the chemistry laboratory would be given one of the following as a first aid

(A) Ice
(B) Petroleum jelly
(C) Gentian violet
(D) Antiseptic

(viii) Which of the following is a natural acid?

(A) Nitric acid
(B) Phosphoric acid
(C) Sulphuric acid
(D) Citric acid

(ix) What is the valency of Group I elements?

(A) 1 (B) 2 (C)3 (d) 4

(x) Which description corresponds to a proton?

(A) Relative mass = 1, charge = 0
(B) Relative mass = 4, charge = + 2
(C) Relative mass = 1, charge = + 1
(D) Relative mass = $\frac{1}{1840}$, charge = -1

QNS	i	ii	iii	iv	v	vi	vii	viii	ix	x
ANSW										

2. Match each item in List A with a correct response in List B by writing its letter below the number of the corresponding item in the table provided.

LIST A	LIST B
(i) Combustion products	A. Fuel, heat and oxygen gas
(ii) A part where the fire extinguisher should be aimed so as to extinguish a fire.	B. Luminous flame
(iii) It has unlimited supply to oxygen	C. Water gas, wood and coal
(iv) Fuels	D. Carbon dioxide, water vapour and energy
(v) It is suitable to extinguish class A fire and fire caused by electrical faults	E. At the top of a fire
	F. Non luminous flame
	G. At the base of a fire
	H. Foam
	I. Halon

LIST A	i	ii	iii	iv	v
LIST B					

SECTION B (70 Marks)

Answer all questions from this section

3. By using one example for each, explain how isotopes are applied in the following areas
   (i) Agriculture

MOCK EXAMINATIONS ARUSHA

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(ii) Industrial and environmental fields

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(iii) Medicine

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(iv) Electricity production

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(v) Archaeology

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4. (a) What are the main safety considerations and perceptions when working with basic chemistry laboratory apparatus and chemicals.

(i) ……
(ii) ……
(iii) ……
(iv) ……

(b)Which reasons will you use to convince your friend that when she/he works in a chemistry laboratory she/he must adhere to safety measures?

(i) ……
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(ii) ……
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(iii)……
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(c)On your field trip, you found two people surrounding a person lying down suspected of shock. How would you advise them on the right way of positioning that victim? Explain the advantage of that positioning.

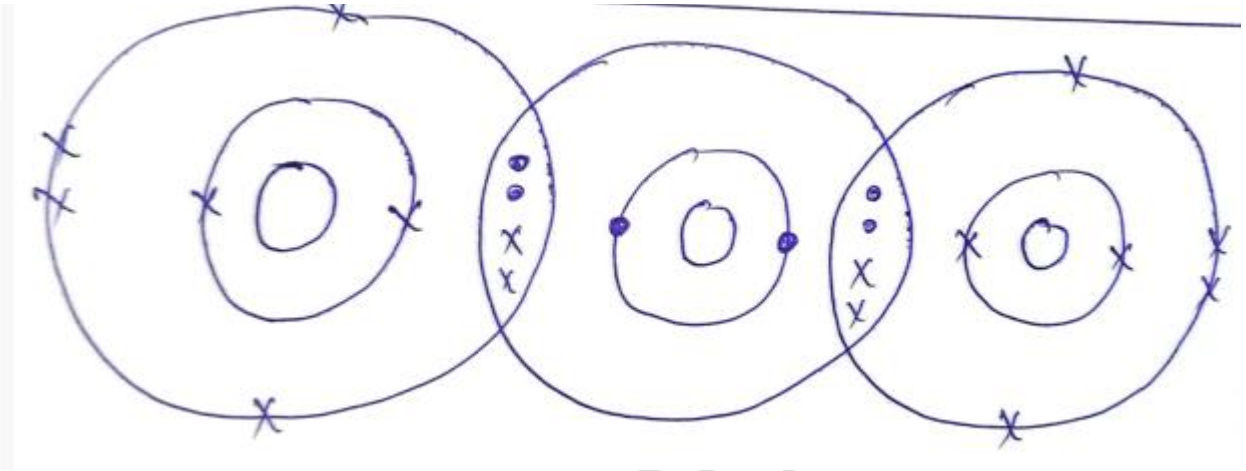
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MOCK EXAMINATIONS ARUSHA

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5. (a) Consider the following molecules of a certain compound, then answer the questions that follows.



Questions

(i) What is the name of the molecule?
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(ii) What is the molecular formula of the compound?
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(iii) What type of bond holds the molecules?
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(iv) Give any other two examples of compound with the same type of bond identified in (a) iii……..

(b) Calculate the oxidation state of the underlined elements in each of the following

Compound

(i) $Pb\underline{S}_2$
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(ii) $\underline{Cu}_2\ SO_4$
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(iii) $H\underline{C}O_3$
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MOCK EXAMINATION

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(iv) $NH_4\underline{O}H$

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6. (a) Draw a well labelled diagram to show the location of sub atomic particles in the atom

(b)State the number of protons, neutrons and electrons in the following atoms

(i) $^{27}_{13}AL$

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(ii) $^{137}_{56}Ba$

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(iii) $^{90}_{38}Sr$

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MOCK EXAMINATIONS ARUSHA

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(iv) $^{235}_{92}U$

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7. (a) A cooker wants to buy a cooking pot the shop keeper after two options, one made from aluminum and the other made from the shiny brittle material. Based on periodic classification, which material is better for cooking pots? Justify your choice.

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(b)Describe properties of metals and explain how do they make metal useful in various applications?

(i) ……..
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(ii) ……
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(iii) ……
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8. (a) Is you have told to highlight the scientific skills that the chemistry must acquire to successfully obtain facts and verify them, what would be your highlights? Write them systematically

(i) ……
(ii) ……
(iii) ……
(iv) ……
(v) ……
(vi) ……..

(b)Give the meaning of the following words as are applied in chemistry

(i) Home care products

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MOCK EXAMINATIONS ARUSHA

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(ii) Biotechnology

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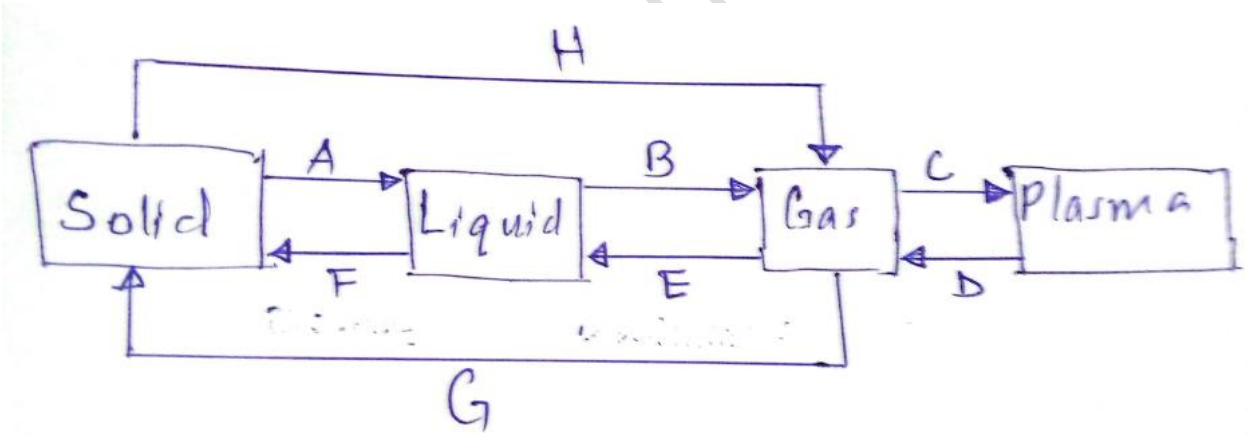
(iii) Agro chemistry

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(iv) Geo – chemists

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9. (a) The diagram below indicates the interconversion of the states a matter, the processes involved in changing one state of matter to another are represented by letter A – H. For each letter write the actual process.



A. ……
B. ……
C. ……
D. ……
E. ……
F. ……...
G. ……..
H. ……

(b)Describe the role of changes of state of matter in water purification

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SECTION C (15 marks)

Answer this question

10. When your group is per forming an experiment in the laboratory and one of the students touches the lose electric wire and get electric shock accidentally, briefly explain how you will render a first Aid to your colleague.

MOCK EXAMINATIONS ARUSHA